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Reflection: Mini Calculator (HW13a)

1. Design Decisions and Rationale

For the Mini Calculator application, I applied the **Separation of Concerns** principle to ensure modularity and maintainability. The application is divided into two primary classes:

- **CalculatorView:** Responsible for the Java Swing-based GUI and event handling.
- **CalculatorLogic:** A decoupled component focused strictly on performing arithmetic operations. This architectural choice isolates the UI logic from the calculation engine, facilitating easier testing and future scalability, such as adding advanced scientific functions without altering the interface code.

2. Main Functionalities

The application features a standard `JTextField` for output display and a responsive grid of buttons. Key functionalities include:

- **Basic Arithmetic:** Support for addition, subtraction, multiplication, division, and modulo operations.
- **Control Mechanisms:** Includes a "Clear" (C) button to reset the state and a "Backspace" (<-) button for precise character correction.
- **Robustness:** The application handles invalid inputs gracefully, such as catching division-by-zero errors and incomplete expressions, preventing potential crashes.

3. Implementation Overview

The GUI was constructed using a `JFrame` with a `BorderLayout` for the display and a `5x4 GridLayout` for the keypad. Event handling is managed via `ActionListener` interfaces attached to each button. Upon triggering the "=" button, the system parses the operands and operators and passes them to the `CalculatorLogic.calculate()` method. I utilized `try-catch` blocks to capture `ArithmeticException` and `NumberFormatException`, ensuring a smooth and error-proof user experience.